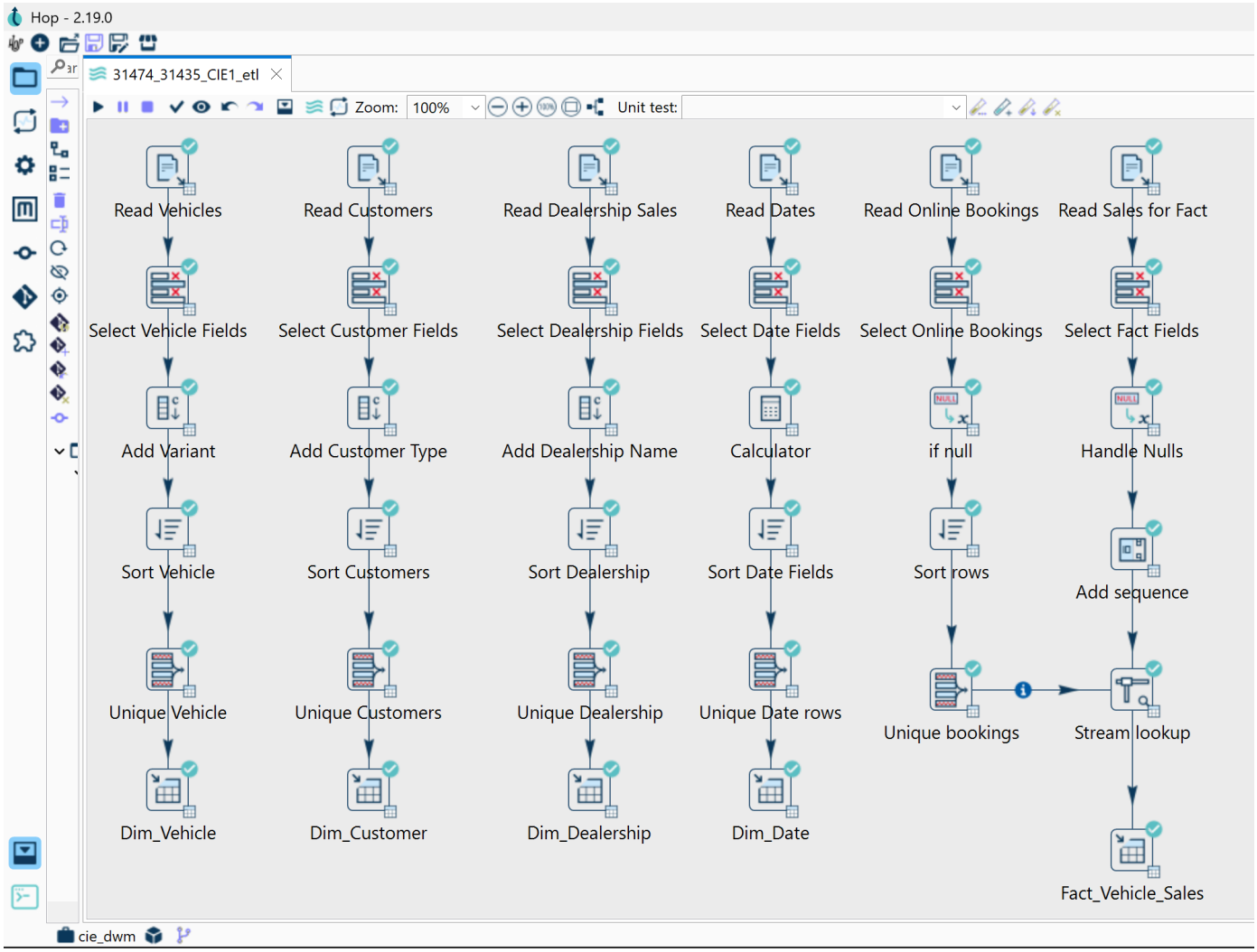
DWM A2

Name: 31474 - Shreeya Tapaswi

Title: Case Study 10: Automobile Sales Retail Industry Data Warehouse

1. ETL Pipeline



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2. Database Before ETL

```
mysql> show tables;
+--------------------------+
| Tables_in_31474_dwm_cie1 |
+--------------------------+
| dim_customer             |
| dim_date                 |
| dim_dealership           |
| dim_vehicle              |
| fact_vehicle_sales       |
+--------------------------+
5 rows in set (0.00 sec)
```

3. Database After ETL

```
mysql> select * from dim_vehicle;
+------------+--------------+----------+-----------+----------+
| Vehicle_ID | Vehicle_Type | Model    | Fuel_Type | Variant  |
+------------+--------------+----------+-----------+----------+
| V001       | SUV          | Creta    | Petrol    | Standard |
| V002       | Sedan        | City     | Petrol    | Standard |
| V003       | SUV          | Nexon    | Electric  | Standard |
| V004       | Hatchback    | Swift    | Petrol    | Standard |
| V005       | SUV          | Seltos   | Diesel    | Standard |
| V006       | SUV          | Hyryder  | Hybrid    | Standard |
| V007       | SUV          | XUV700   | Diesel    | Standard |
| V008       | Hatchback    | i20      | Petrol    | Standard |
| V009       | SUV          | Elevate  | Petrol    | Standard |
| V010       | SUV          | Nexon EV | Electric  | Standard |
| V011       | SUV          | Brezza   | Petrol    | Standard |
| V012       | SUV          | Sonet    | Diesel    | Standard |
| V013       | Sedan        | Camry    | Hybrid    | Standard |
| V014       | Sedan        | Slavia   | Petrol    | Standard |
| V015       | SUV          | Taigun   | Petrol    | Standard |
| V016       | Hatchback    | Punch    | Petrol    | Standard |
| V017       | Sedan        | Verna    | Petrol    | Standard |
| V018       | SUV          | Thar     | Diesel    | Standard |
| V019       | SUV          | Kiger    | Petrol    | Standard |
| V020       | Hatchback    | Comet EV | Electric  | Standard |
+------------+--------------+----------+-----------+----------+
20 rows in set (0.00 sec)
```

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```
mysql> select * from dim_date;
+---------+---------------------+------+-------+------+---------+
| Date_ID | Full_Date           | Year | Month | Day  | Quarter |
+---------+---------------------+------+-------+------+---------+
| DT001   | 2026-01-05 00:00:00 | 2026 |     1 |    5 |       1 |
| DT002   | 2026-01-12 00:00:00 | 2026 |     1 |   12 |       1 |
| DT003   | 2026-01-24 00:00:00 | 2026 |     1 |   24 |       1 |
| DT004   | 2026-02-03 00:00:00 | 2026 |     2 |    3 |       1 |
| DT005   | 2026-02-14 00:00:00 | 2026 |     2 |   14 |       1 |
| DT006   | 2026-02-26 00:00:00 | 2026 |     2 |   26 |       1 |
| DT007   | 2026-03-08 00:00:00 | 2026 |     3 |    8 |       1 |
| DT008   | 2026-03-18 00:00:00 | 2026 |     3 |   18 |       1 |
| DT009   | 2026-04-02 00:00:00 | 2026 |     4 |    2 |       2 |
| DT010   | 2026-04-15 00:00:00 | 2026 |     4 |   15 |       2 |
| DT011   | 2026-04-27 00:00:00 | 2026 |     4 |   27 |       2 |
| DT012   | 2026-05-06 00:00:00 | 2026 |     5 |    6 |       2 |
| DT013   | 2026-05-19 00:00:00 | 2026 |     5 |   19 |       2 |
| DT014   | 2026-06-01 00:00:00 | 2026 |     6 |    1 |       2 |
| DT015   | 2026-06-16 00:00:00 | 2026 |     6 |   16 |       2 |
| DT016   | 2026-07-04 00:00:00 | 2026 |     7 |    4 |       3 |
| DT017   | 2026-07-19 00:00:00 | 2026 |     7 |   19 |       3 |
| DT018   | 2026-08-07 00:00:00 | 2026 |     8 |    7 |       3 |
| DT019   | 2026-08-21 00:00:00 | 2026 |     8 |   21 |       3 |
| DT020   | 2026-09-10 00:00:00 | 2026 |     9 |   10 |       3 |
+---------+---------------------+------+-------+------+---------+
20 rows in set (0.00 sec)
```

```
mysql> select * from fact_vehicle_sales;
+---------+-------------+------------+-----------+---------+--------------+----------+----------------+
| Sale_ID | Customer_ID | Vehicle_ID | Dealer_ID | Date_ID | Sales_Amount | Quantity | Booking_Amount |
+---------+-------------+------------+-----------+---------+--------------+----------+----------------+
| S001    | C001        | V001       | D001      | DT001   |      1425000 |        1 |           5000 |
| S002    | C002        | V002       | D002      | DT002   |      1220000 |        1 |           6000 |
| S003    | C003        | V003       | D003      | DT003   |      1080000 |        1 |           7000 |
| S004    | C004        | V004       | D004      | DT004   |       835000 |        1 |           8000 |
| S005    | C005        | V005       | D005      | DT005   |      1510000 |        1 |           5000 |
| S006    | C006        | V006       | D006      | DT006   |      1775000 |        1 |           6000 |
| S007    | C007        | V007       | D007      | DT007   |      2070000 |        1 |           7000 |
| S008    | C008        | V008       | D008      | DT008   |       930000 |        1 |           8000 |
| S009    | C009        | V009       | D009      | DT009   |      1635000 |        1 |           5000 |
| S010    | C010        | V010       | D010      | DT010   |      1560000 |        1 |           6000 |
| S011    | C011        | V011       | D011      | DT011   |      1175000 |        1 |           7000 |
| S012    | C012        | V012       | D012      | DT012   |      1020000 |        1 |           8000 |
| S013    | C013        | V013       | D013      | DT013   |      4580000 |        1 |           5000 |
| S014    | C014        | V014       | D014      | DT014   |      1735000 |        1 |           6000 |
| S015    | C015        | V015       | D015      | DT015   |      1810000 |        1 |           7000 |
| S016    | C016        | V016       | D016      | DT016   |       775000 |        1 |           8000 |
| S017    | C017        | V017       | D017      | DT017   |      1420000 |        1 |           5000 |
| S018    | C018        | V018       | D018      | DT018   |      1730000 |        1 |           6000 |
| S019    | C019        | V019       | D019      | DT019   |       885000 |        1 |           7000 |
| S020    | C020        | V020       | D020      | DT020   |       810000 |        1 |           8000 |
+---------+-------------+------------+-----------+---------+--------------+----------+----------------+
20 rows in set (0.00 sec)
```

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OLAP Operations

1. Roll-Up (Moving from a detailed level to a higher/summarized level)

- Query 1.1: Sales Roll-Up by Region (Dealership hierarchy roll-up)

```
mysql> SELECT d.Region, SUM(f.Sales_Amount) AS Total_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_dealership d ON f.Dealer_ID = d.Dealership_ID
    -> GROUP BY d.Region;
+----------------+-------------+
| Region         | Total_Sales |
+----------------+-------------+
| Maharashtra    |    10825000 |
| Gujarat        |     2845000 |
| Karnataka      |     2660000 |
| Delhi          |     3370000 |
| Telangana      |     1175000 |
| Kerala         |     1020000 |
| Rajasthan      |     4580000 |
| Chandigarh     |     1810000 |
| Uttar Pradesh  |      885000 |
| Madhya Pradesh |      810000 |
+----------------+-------------+
10 rows in set (0.00 sec)
```

- Query 1.2: Sales Roll-Up by Quarter (Date hierarchy roll-up)

```
mysql> SELECT dt.Quarter, SUM(f.Sales_Amount) AS Total_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_date dt ON f.Date_ID = dt.Date_ID
    -> GROUP BY dt.Quarter;
+---------+-------------+
| Quarter | Total_Sales |
+---------+-------------+
|       1 |    10845000 |
|       2 |    13515000 |
|       3 |     5620000 |
+---------+-------------+
3 rows in set (0.00 sec)
```

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2. Drill-Down (Moving from a summary level to a more detailed level)

- **Query 2.1: Drill-Down from Year to Month**

```
mysql> SELECT dt.Year, dt.Month, SUM(f.Sales_Amount) AS Total_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_date dt ON f.Date_ID = dt.Date_ID
    -> GROUP BY dt.Year, dt.Month
    -> ORDER BY dt.Year, dt.Month;
+------+-------+-------------+
| Year | Month | Total_Sales |
+------+-------+-------------+
| 2026 |     1 |     3725000 |
| 2026 |     2 |     4120000 |
| 2026 |     3 |     3000000 |
| 2026 |     4 |     4370000 |
| 2026 |     5 |     5600000 |
| 2026 |     6 |     3545000 |
| 2026 |     7 |     2195000 |
| 2026 |     8 |     2615000 |
| 2026 |     9 |      810000 |
+------+-------+-------------+
9 rows in set (0.00 sec)
```

- **Query 2.2: Drill-Down into Vehicle Model level (from Vehicle Type)**

```
mysql> SELECT v.Vehicle_Type, v.Model, SUM(f.Sales_Amount) AS Total_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_vehicle v ON f.Vehicle_ID = v.Vehicle_ID
    -> GROUP BY v.Vehicle_Type, v.Model
    -> ORDER BY v.Vehicle_Type, v.Model;
+--------------+----------+-------------+
| Vehicle_Type | Model    | Total_Sales |
+--------------+----------+-------------+
| Hatchback    | Comet EV |      810000 |
| Hatchback    | i20      |      930000 |
| Hatchback    | Punch    |      775000 |
| Hatchback    | Swift    |      835000 |
| Sedan        | Camry    |     4580000 |
| Sedan        | City     |     1220000 |
| Sedan        | Slavia   |     1735000 |
| Sedan        | Verna    |     1420000 |
| SUV          | Brezza   |     1175000 |
| SUV          | Creta    |     1425000 |
| SUV          | Elevate  |     1635000 |
| SUV          | Hyryder  |     1775000 |
| SUV          | Kiger    |      885000 |
| SUV          | Nexon    |     1080000 |
| SUV          | Nexon EV |     1560000 |
| SUV          | Seltos   |     1510000 |
| SUV          | Sonet    |     1020000 |
| SUV          | Taigun   |     1810000 |
| SUV          | Thar     |     1730000 |
| SUV          | XUV700   |     2070000 |
+--------------+----------+-------------+
20 rows in set (0.00 sec)
```

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3. Slice (Fixing one dimension to isolate a specific sub-cube slice)

- **Query 3.1: Slice for 'SUV' Vehicle Type only**

```
mysql> SELECT v.Model, SUM(f.Sales_Amount) AS Total_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_vehicle v ON f.Vehicle_ID = v.Vehicle_ID
    -> WHERE v.Vehicle_Type = 'SUV'
    -> GROUP BY v.Model;
+----------+-------------+
| Model    | Total_Sales |
+----------+-------------+
| Creta    |     1425000 |
| Nexon    |     1080000 |
| Seltos   |     1510000 |
| Hyryder  |     1775000 |
| XUV700   |     2070000 |
| Elevate  |     1635000 |
| Nexon EV |     1560000 |
| Brezza   |     1175000 |
| Sonet    |     1020000 |
| Taigun   |     1810000 |
| Thar     |     1730000 |
| Kiger    |      885000 |
+----------+-------------+
12 rows in set (0.00 sec)
```

- **Query 3.2: Slice for 'Maharashtra' Region only**

```
mysql> SELECT d.Region, f.Sale_ID, f.Sales_Amount
    -> FROM fact_vehicle_sales f
    -> JOIN dim_dealership d ON f.Dealer_ID = d.Dealership_ID
    -> WHERE d.Region = 'Maharashtra';
+-------------+---------+--------------+
| Region      | Sale_ID | Sales_Amount |
+-------------+---------+--------------+
| Maharashtra | S001    |      1425000 |
| Maharashtra | S002    |      1220000 |
| Maharashtra | S003    |      1080000 |
| Maharashtra | S004    |       835000 |
| Maharashtra | S005    |      1510000 |
| Maharashtra | S006    |      1775000 |
| Maharashtra | S010    |      1560000 |
| Maharashtra | S017    |      1420000 |
+-------------+---------+--------------+
8 rows in set (0.00 sec)
```

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4. Dice (Filtering across multiple dimensions simultaneously)

- **Query 4: Dice by Hatchback vehicles and Quarter 1 sales**

```
mysql> SELECT v.Model, dt.Quarter, SUM(f.Sales_Amount) AS Total_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_vehicle v ON f.Vehicle_ID = v.Vehicle_ID
    -> JOIN dim_date dt ON f.Date_ID = dt.Date_ID
    -> WHERE v.Vehicle_Type = 'Hatchback'
    ->   AND dt.Quarter = 1
    -> GROUP BY v.Model, dt.Quarter;
+-------+---------+-------------+
| Model | Quarter | Total_Sales |
+-------+---------+-------------+
| Swift |       1 |      835000 |
| i20   |       1 |      930000 |
+-------+---------+-------------+
2 rows in set (0.00 sec)
```

5. Pivot / Rotation (Rotating the axes to view data from a different perspective, often using conditional aggregates)

- **Query 5.1: Pivot Vehicle Types against Fuel Types (Count or Sales Sum)**

```
mysql> SELECT
    ->     v.Vehicle_Type,
    ->     SUM(CASE WHEN v.Fuel_Type = 'Petrol' THEN f.Sales_Amount ELSE 0 END) AS Petrol_Sales,
    ->     SUM(CASE WHEN v.Fuel_Type = 'Electric' THEN f.Sales_Amount ELSE 0 END) AS Electric_Sales,
    ->     SUM(CASE WHEN v.Fuel_Type = 'Diesel' THEN f.Sales_Amount ELSE 0 END) AS Diesel_Sales
    -> FROM fact_vehicle_sales f
    -> JOIN dim_vehicle v ON f.Vehicle_ID = v.Vehicle_ID
    -> GROUP BY v.Vehicle_Type;
+--------------+--------------+----------------+--------------+
| Vehicle_Type | Petrol_Sales | Electric_Sales | Diesel_Sales |
+--------------+--------------+----------------+--------------+
| SUV          |      6930000 |        2640000 |      6330000 |
| Sedan        |      4375000 |              0 |            0 |
| Hatchback    |      2540000 |         810000 |            0 |
+--------------+--------------+----------------+--------------+
3 rows in set (0.00 sec)
```

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- **Query 5.2: Pivot Quarters against Vehicle Types for Sales Volume**

```
mysql> SELECT
    ->     v.Vehicle_Type,
    ->     SUM(CASE WHEN dt.Quarter = 1 THEN f.Quantity ELSE 0 END) AS Q1_Qty,
    ->     SUM(CASE WHEN dt.Quarter = 2 THEN f.Quantity ELSE 0 END) AS Q2_Qty,
    ->     SUM(CASE WHEN dt.Quarter = 3 THEN f.Quantity ELSE 0 END) AS Q3_Qty
    -> FROM fact_vehicle_sales f
    -> JOIN dim_vehicle v ON f.Vehicle_ID = v.Vehicle_ID
    -> JOIN dim_date dt ON f.Date_ID = dt.Date_ID
    -> GROUP BY v.Vehicle_Type;
+--------------+--------+--------+--------+
| Vehicle_Type | Q1_Qty | Q2_Qty | Q3_Qty |
+--------------+--------+--------+--------+
| SUV          |      5 |      5 |      2 |
| Sedan        |      1 |      2 |      1 |
| Hatchback    |      2 |      0 |      2 |
+--------------+--------+--------+--------+
3 rows in set (0.00 sec)
```